# AR/VR Workshop
# **Introduction**

Phillip G. Bradford

University of Connecticut

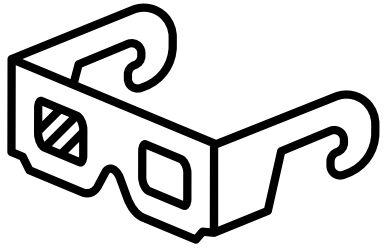
# Full select demo – how you can do …

Basic-Mars-Landing

# Module demo highlights: 3D image

# Outline

Overview

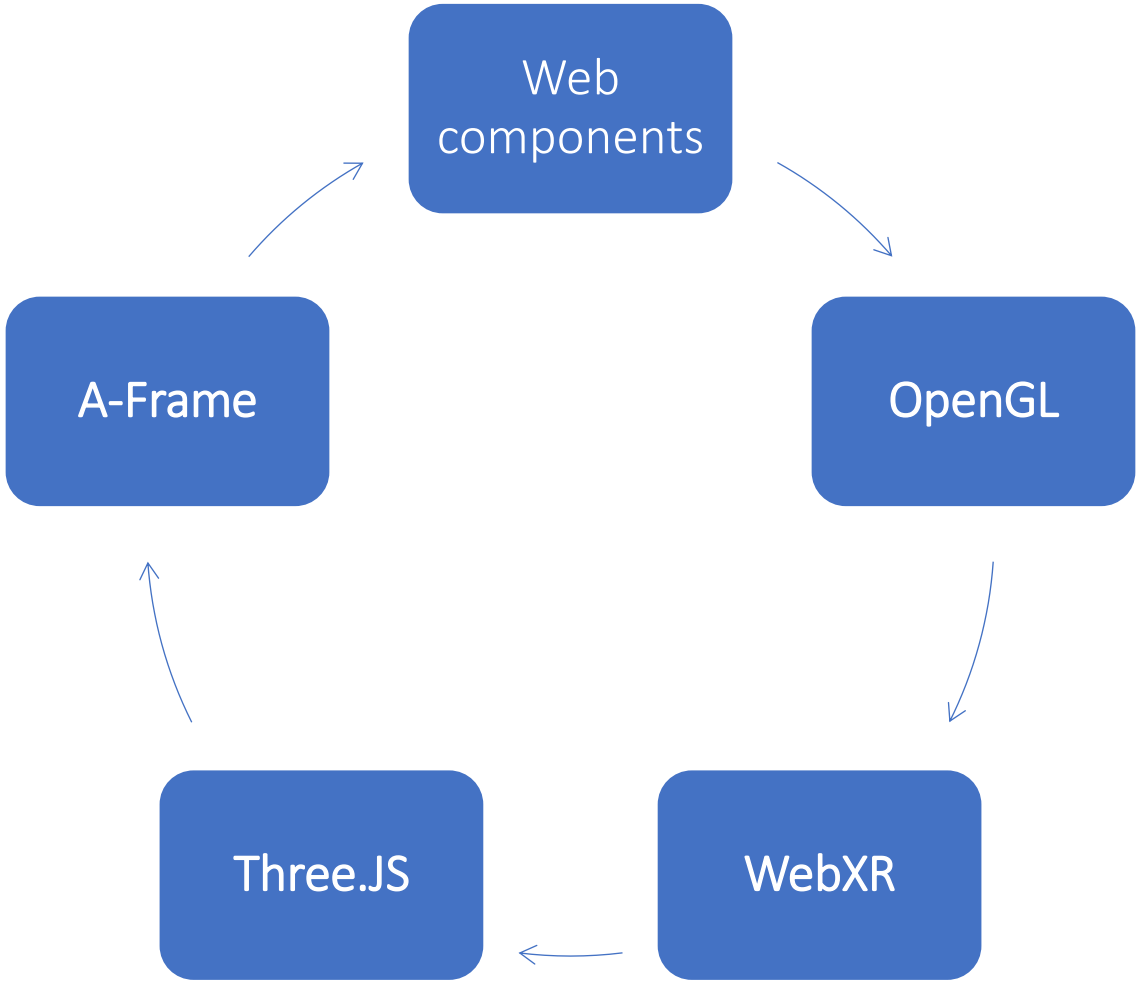


No installations needed

Github pages

Our first VR/AR examples

# High level view and learning path



# A-frame

3D virtual reality system built by Mozilla

**A-frame** is a component system built on **Three.js**

**Three.js** is built on **WebXL** which is built on **WebGL**

**A-frame** is HTML that uses web-components

Declarative – tell it what you want to see,

it figures out how to do it

# No installation VR

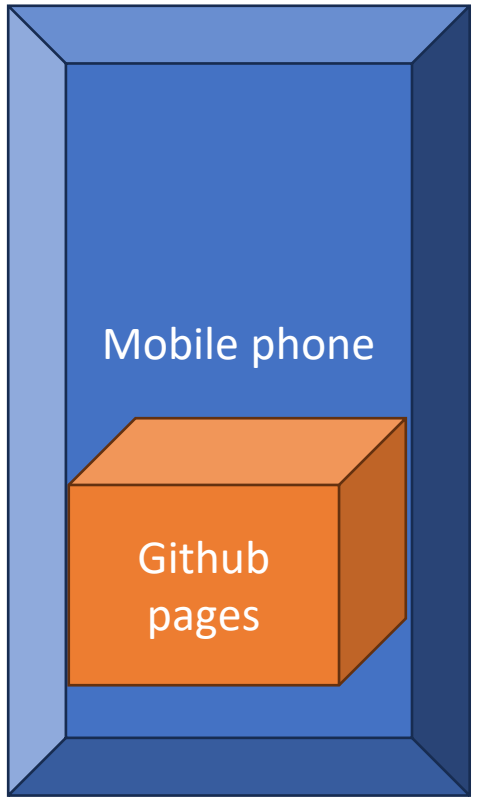
Easy

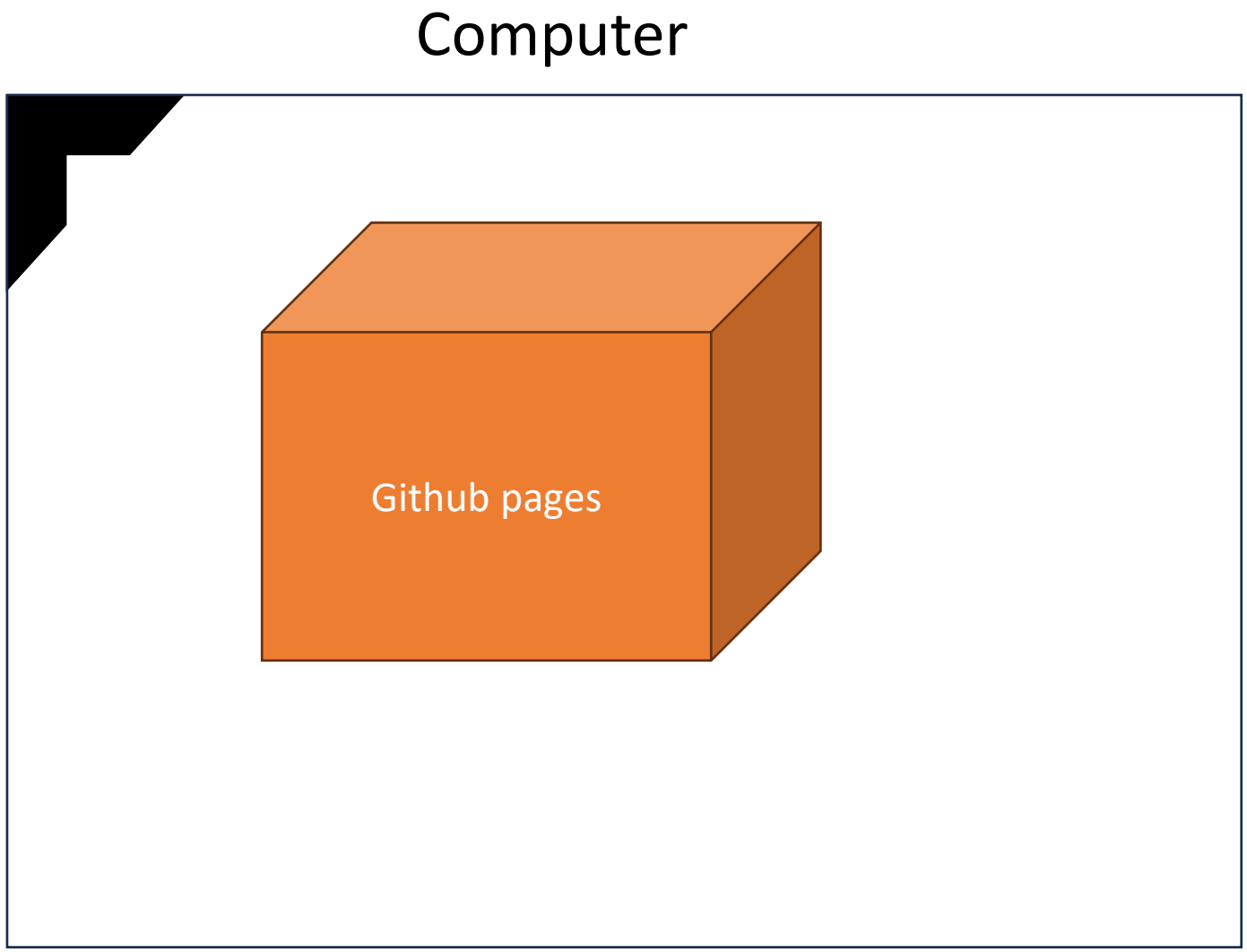
Only Python3 installed

Goto **wonder-phil.github.io** or YOUR_USER. github.io

Clone https://github.com/wonder-phil/wonder-phil.github.io.git

# Overview: wonder-phil.github.io



# Laptop view

git clone https://github.com/wonder-phil/wonder-phil.github.io.git

cd **wonder-phil.github.io**

**Term1> python –m http.server**

In a browser go to **http://localhost:8000/**

**Term2> code .**

# Basic HTML

```
<html>
  <head>

  </head>
  <body>

  </body>
</html>
```

# index.html and not-index.html

```
<html>
<head>
   <meta charset="utf-8">
   <title>Hello A-Frame Example</title>
    <script src="https://aframe.io/releases/1.4.1/aframe.min.js"></script>
  </head>
<body>
   <a-scene>
    <a-sky color="#0000EE"></a-sky>
    <a-text
      font="kelsonsans"
      color = "#FFFFFF"
      value="HELLO A-Frame!!!"
      width="6"
      position="0 -1 -5"
    ></a-text>
  </a-scene>
  </body>
</html>
```

```
<a-scene>
    <a-sky color="#0000EE“>
     </a-sky>
    <a-text
      font="kelsonsans"
      color = "#FFFFFF"
      value="HELLO A-Frame!!!"
      width="6"
      position="0 -1 -5"
    ></a-text>
  </a-scene>
```

# VR glasses steps

On your phone go to your github pages

**wonder-phil.github.io**

Use VR glasses to view the project on your phone

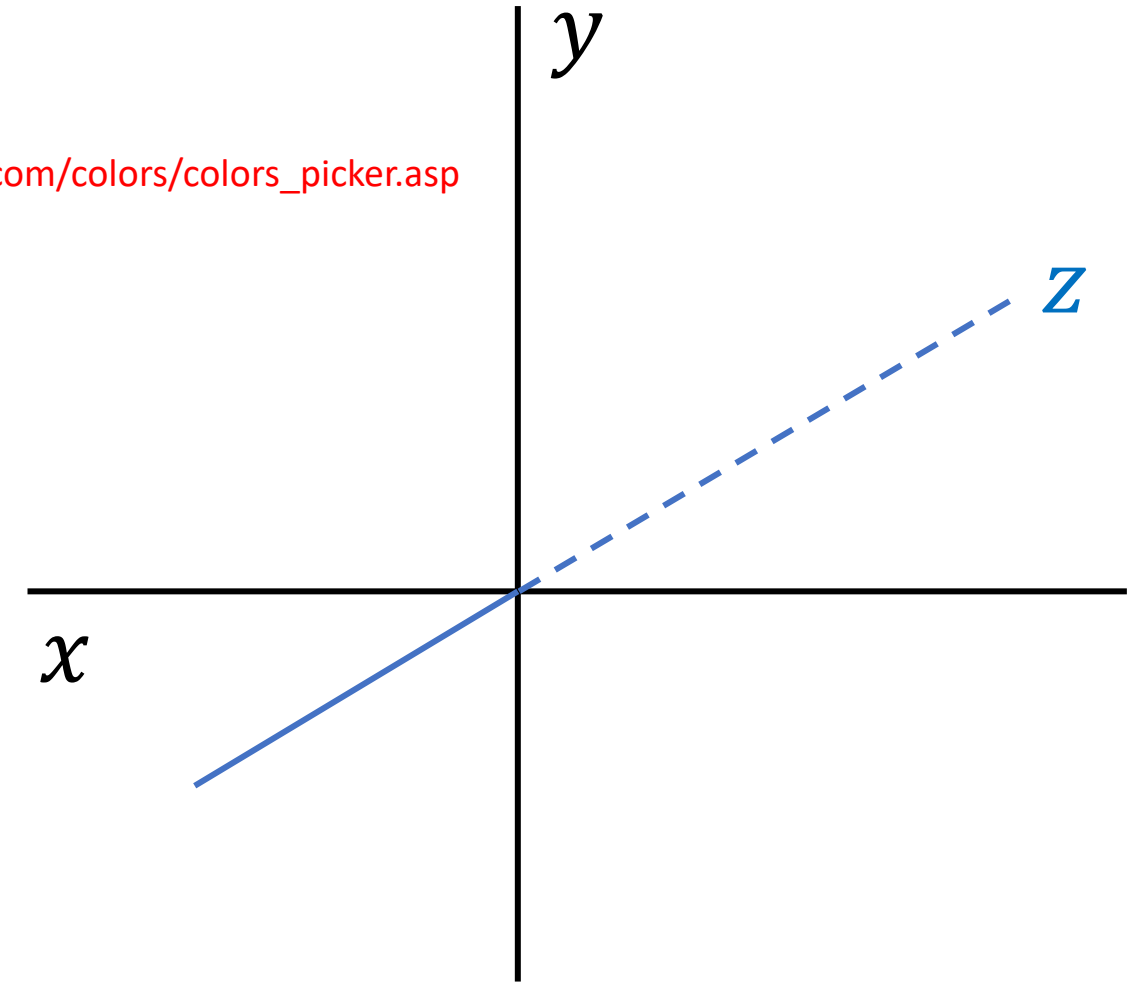
# simple-sky.html

```
<a-scene>
    <a-sky color="#0105EF">
    </a-sky>

    <a-text
     font="kelsonsans"
     color = "#FFFFFF"
     value="HELLO A-Frame!!!"
     width="6"
     position="0  -1  -5“
    ></a-text>

  </a-scene>
```

Select a color at https://www.w3schools.com/colors/colors_picker.asp

Note: **x-horizontal, y-vertical, and z-depth***

*this translates to 1 meter below the origin and 5 meters into the page



# simple-sky.html: changing the view

```
<a-scene>
    <a-sky color="#0105EF">
    </a-sky>
    <a-text
     font="kelsonsans"
     color = "#FFFFFF"
     value="HELLO A-Frame!!!"
     width="6"
     position="0   -1   -5"
    ></a-text>
  </a-scene>
```

Replace the current index.html on glitch with not-index.html
Simple-sky.html

← a-scene now looks like this

# Putting it all together

You have now created virtual reality text on a 3D background

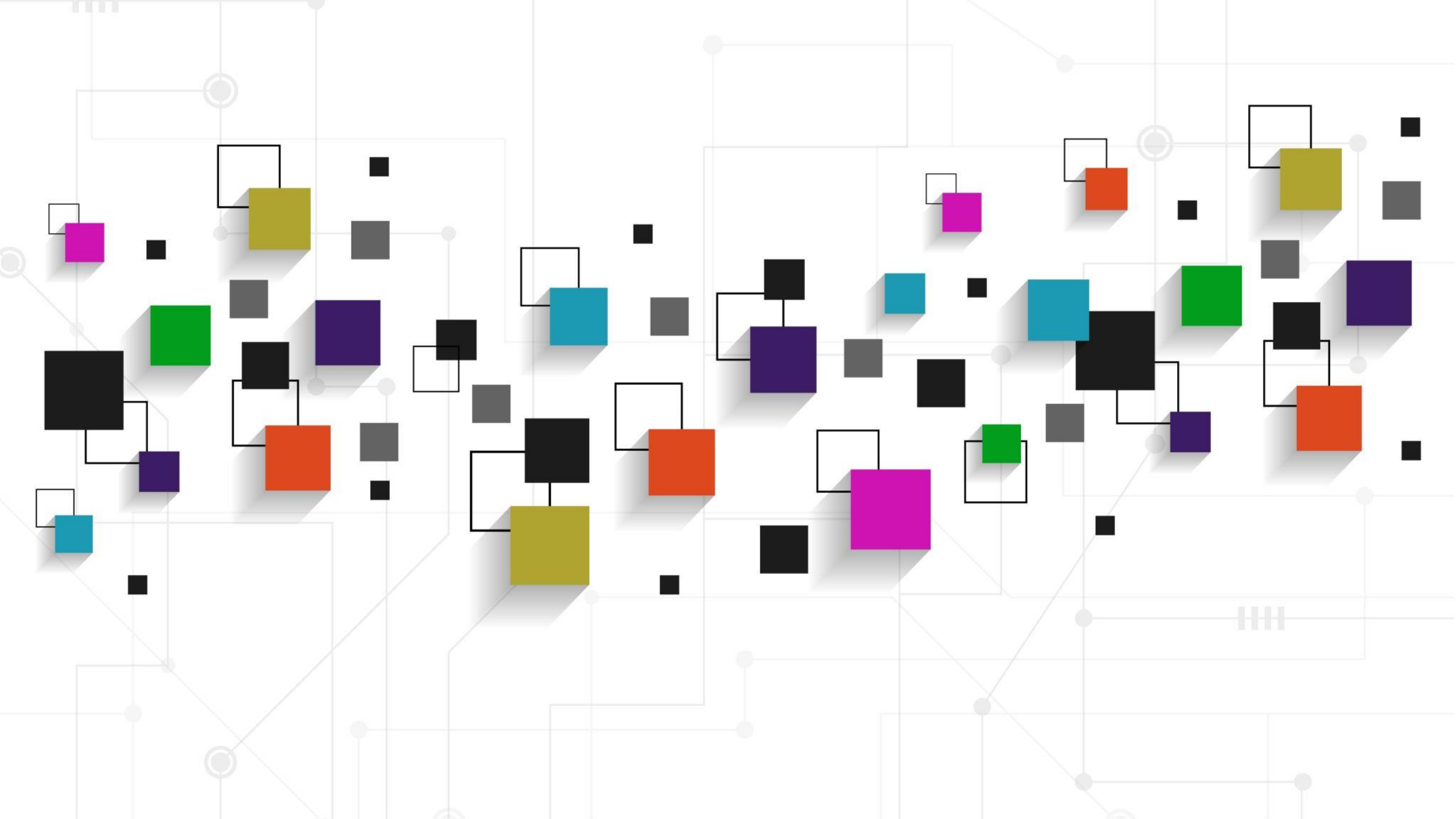
We  can also

1. Change background – to the nebula

What if it doesn’t work?  TROUBLESHOOTING!

2. Move text around in the nebula

What if it doesn’t work?  TROUBLESHOOTING!



# Getting a-frame from aframe.io

```
<head>
   <script src="https://aframe.io/releases/1.7.1/aframe.min.js">
   </script>
</head>
```

Or to run outside of Glitch.com to help TROUBLE SHOOTING

```
 <body>
  <script src="../js/aframe.io_releases_1.7.1_aframe.min.js">
  </script>
```

# TROUBLESHOOTING!

CTRL-ALT-I in browser for a-frame debugging

CMD-Option-I in browser for a-frame debugging

SHIFT-CTRL-J in browser for Chrome/FireFox debugging

CTRL-O in browser to load a file